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| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The list is sorted
- c. The search continues from the beginning
- d. The search stops

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Bubble search
- c. Merge search
- d. Linear search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 10 ✓
- b. 500
- c. 100
- d. 1000

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Skipping every second element
- c. Sorting the list before searching
- d. Dividing the list in half repeatedly

Question **5**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the list contains duplicate elements
- c. When the list is sorted
- d. When the search algorithm finishes

Question **6**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Hash search
- c. Binary search
- d. Both (1) & (3)

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There must be a mechanism to delete and/or insert elements in the list ✓
- b. Number values should only be present

  Number values should only be present
- c. The list must be sorted
- d. There should be direct access to the middle element in any sublist

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The search is considered unsuccessful ✓
c. The first element is returned
d. The last element is returned

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O(1)
c. O(n log n)
d. O👎

Question **10**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. There must be a mechanism to access middle element directly
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **12**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The search stops
c. The lower middle element is chosen as the middle element ✗
d. The higher middle element is chosen as the middle element

Question **13**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Hash search
c. Binary search ✓
d. Sequential search

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search is more efficient than binary search on large lists.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Binary search
c. Linear search ✓
d. Hashing